

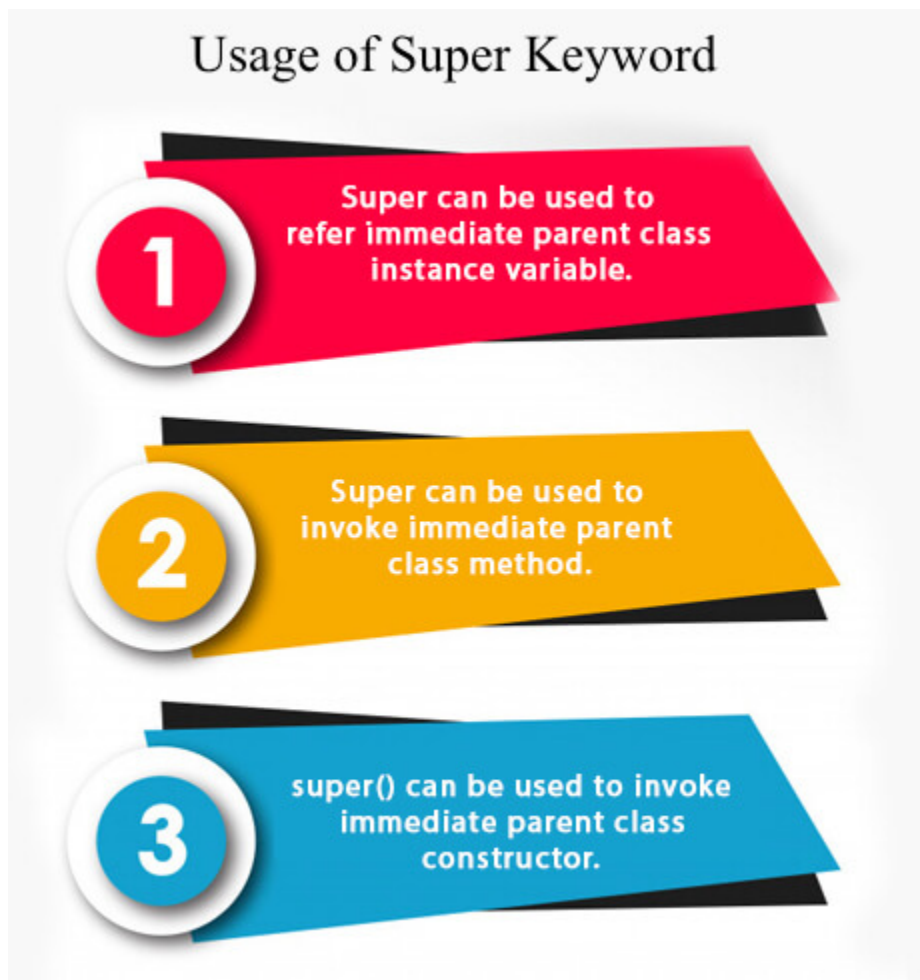
Super Keyword in Java

The **super** keyword in Java is a reference variable which is used to refer immediate parent class object.

Whenever you create the instance of subclass, an instance of parent class is created implicitly which is referred by super reference variable.

Usage of Java super Keyword

1. super can be used to refer immediate parent class instance variable.
2. super can be used to invoke immediate parent class method.
3. super() can be used to invoke immediate parent class constructor.



1) super is used to refer immediate parent class instance variable.

We can use super keyword to access the data member or field of parent class. It is used if parent class and child class have same fields.

```
class Animal{
String color="white";
}
class Dog extends Animal{
String color="black";
void printColor(){
System.out.println(color);//prints color of Dog class
System.out.println(super.color);//prints color of Animal class
}
}
class TestSuper1{
public static void main(String args[]){
Dog d=new Dog();
d.printColor();
}}
```

Test it Now

Output:

```
black
white
```

In the above example, Animal and Dog both classes have a common property color. If we print color property, it will print the color of current class by default. To access the parent property, we need to use super keyword.

2) super can be used to invoke parent class method

The super keyword can also be used to invoke parent class method. It should be used if subclass contains the same method as parent class. In other words, it is used if method is overridden.

```
class Animal{
```

```
void eat(){System.out.println("eating...");}
}
class Dog extends Animal{
void eat(){System.out.println("eating bread...");}
void bark(){System.out.println("barking...");}
void work(){
super.eat();
bark();
}
}
class TestSuper2{
public static void main(String args[]){
Dog d=new Dog();
d.work();
}}
```

Test it Now

Output:

```
eating...
barking...
```

In the above example Animal and Dog both classes have eat() method if we call eat() method from Dog class, it will call the eat() method of Dog class by default because priority is given to local.

To call the parent class method, we need to use super keyword.

3) super is used to invoke parent class constructor.

The super keyword can also be used to invoke the parent class constructor. Let's see a simple example:

```
class Animal{
Animal(){System.out.println("animal is created");}
}
class Dog extends Animal{
```

```

Dog(){
    super();
    System.out.println("dog is created");
}
}

class TestSuper3{
    public static void main(String args[]){
        Dog d=new Dog();
    }
}

```

Test it Now

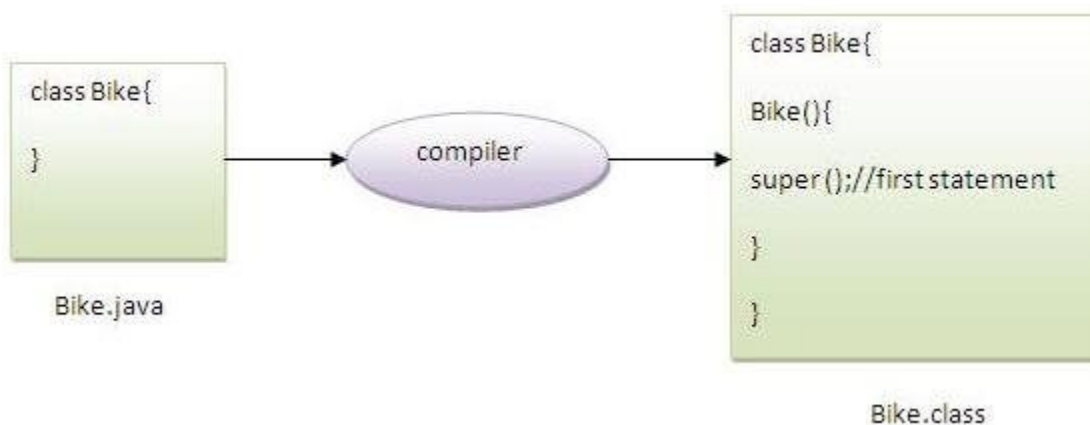
Output:

```

animal is created
dog is created

```

Note: super() is added in each class constructor automatically by compiler if there is no super() or this().



As we know well that default constructor is provided by compiler automatically if there is no constructor. But, it also adds super() as the first statement.

Another example of super keyword where super() is provided by the compiler implicitly.

```
class Animal{
Animal(){System.out.println("animal is created");}
}
class Dog extends Animal{
Dog(){
System.out.println("dog is created");
}
}
class TestSuper4{
public static void main(String args[]){
Dog d=new Dog();
}}
```

Test it Now

Output:

```
animal is created
dog is created
```

super example: real use

Let's see the real use of super keyword. Here, Emp class inherits Person class so all the properties of Person will be inherited to Emp by default. To initialize all the property, we are using parent class constructor from child class. In such way, we are reusing the parent class constructor.

```
class Person{
int id;
String name;
Person(int id,String name){
this.id=id;
this.name=name;
}
}
class Emp extends Person{
float salary;
```

```
Emp(int id,String name,float salary){
    super(id,name);//reusing parent constructor
    this.salary=salary;
}
void display(){System.out.println(id+" "+name+" "+salary);}
}
class TestSuper5{
    public static void main(String[] args){
        Emp e1=new Emp(1,"ankit",45000f);
        e1.display();
    }
}
```

Test it Now

Output:

```
1 ankit 45000
```

Super keyword MCQ

1. Which of the following is a correct usage of the super keyword to call a superclass constructor?

1. super();
2. super.class();
3. superclass();
4. super.constructor();



Show Answer

Workspace

2. When should you use the super keyword to call a superclass method?

1. When overriding a method in a subclass
2. When defining a new method in a subclass
3. When creating an instance of a superclass
4. When defining a static method

[Show Answer](#)[Workspace](#)

3. In which scenario is the super keyword used to access a superclass variable?

1. When the superclass variable name is shadowed by a subclass variable
2. When the superclass variable is private
3. When the subclass does not have access to the superclass variable
4. When initializing static variables

[Show Answer](#)[Workspace](#)

4. What is the primary purpose of the super keyword in a subclass constructor?

1. To ensure that the superclass is properly initialized
2. To call the subclass constructor
3. To define a new method in the subclass
4. To override a superclass method

[Show Answer](#)[Workspace](#)

5. Which of the following statements is true regarding the super keyword?

1. It can only be used in static methods
2. It can be used to access private members of the superclass
3. It is used to refer to the immediate parent class object
4. It can be used to define new variables in the subclass

[Show Answer](#)[Workspace](#)[Next Topic](#)[Instance Initializer Block](#)[← prev](#)[next →](#)

Learn Important Tutorial



Python



Java



Javascript



HTML



Database



PHP



C++



React

B.Tech / MCA



DBMS



DAA

Data
Structures



Operating
System



Computer
Network



Compiler
Design



Computer
Organization



Discrete
Mathematics



Ethical
Hacking



Computer
Graphics



Web
Technology



Software
Engineering



Cyber



Automata



C
Programming



C++



Java



.Net



Python



Programs



Control
System



Data
Warehouse

Preparation



Verbal
Ability

F 3



Interview
Questions



Company
Questions